NAME_____

MEDGAR EVERS COLLEGE
The City University of New York

CS281 - 001: Database Systems
Spring 2026

Midterm Exam

Scoring

Question	Points Earned	Max Points
1.		5
2.		5
3.		5
4.		5
5.		5
6.		5
7.		5
8.		5
9.		5
10.		5
11.		5
12.		5
13.		5
14.		5
15.		5
16.		5
17.		20
EC 1.		10
EC 2.		10
EC 3.		5
Total		**100**

- No electronic devices.
- Leaving the classroom at any point during the exam is discouraged. If a need to leave class arises, cellphones must be left behind.

PASSWORD: 4T0M1C1TY

1. Define the term **relation.**
2. Define the term **tuple.**
3. Define the term **attribute.**
4. What is a **primary key**?
5. With regard to transaction management, what is **atomicity**?
6. Convert the following SQL query into a relational-algebra expression.

```
SELECT name, salary
FROM instructor
WHERE salary < 80000;
```

7. Convert the following SQL query into a relational-algebra expression.

```
SELECT takes.course_id, title, credits, sec_id, semester, year, grade, dept_name
FROM takes
JOIN course
ON takes.course_id = course.course_id
WHERE ID = "12345";
```

8. Convert the following SQL query into a relational-algebra expression.

```
SELECT student.name, instructor.name
FROM student
JOIN advisor
ON student.ID = s_id
JOIN instructor
ON i_id = instructor.ID;
```

9. Convert the following relational-algebra expression into an SQL query.

$\sigma_{\text{dept_name} = \text{“Comp. Sci.”} \wedge \text{credits} = 3}(\text{course})$

10. Convert the following relational-algebra expression into an SQL query.

$\Pi_{\text{name, salary, course_id}}(\text{instructor} \bowtie_{\text{instructor.ID} = \text{teaches.ID}} \text{teaches}))$

11. Convert the following relational-algebra expression into an SQL query.

$\Pi_{\text{course_id, title, grade}}(\sigma_{\text{takes.course_id} = \text{course.course_id}}((\text{student} \bowtie_{\text{student.ID} = \text{takes.ID}} \text{takes}) \times \text{course}))$

12. Find the ID and name of each student who has not taken any course offered before 2018.

13. Show all 300 level courses. Attributes include: course_id, title, dept_name, credits, semester, year, building, room_number

14. Use an aggregate function to show the total instructor salary per department. Show the department name, and the amount renamed as *payroll*.

15. Add a new record to the course table called "Computer Statistics" whose id is "CS-237". This is a 4 credit computer science course.

16. DELETE all introductory courses. Any course with 101 in the course_id or Intro in the title is considered an introductory course.

17. Write SQL DDL corresponding to the schema in Figure 3.17. Make any reasonable assumptions about data types, and be sure to declare primary and foreign keys.

    person (driver_id, name, address)
    car (license_plate, model, year)
    accident (report_number, year, location)
    owns (driver_id, license_plate)
    participated (report_number, license_plate, driver_id, damage_amount)

18. Using a nested query, find the ID and name of each student who has taken at least one Comp. Sci. course; make sure there are no duplicate names in the result.

19. Draw the relational schema diagram for the database from question 17.

20. Who was the real villain?

QUESTION 19: